ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ASSIGNMENT – 8

Apply Regularization to Avoid Overfitting

Dataset: Scikit-learn Diabetes Dataset

1. Aim

To apply regularization techniques to regression models in order to control model complexity, reduce overfitting and compare Linear, Ridge and Lasso Regression.

2. Objectives

- Understand overfitting and model complexity.
- Study L1 and L2 regularization.
- Apply Ridge and Lasso Regression.
- Use cross-validation to select regularization strength.
- Compare regularized models with ordinary Linear Regression.
- Analyze coefficient shrinkage and prediction performance.

3. Dataset Description

The Diabetes dataset contains 442 observations, 10 independent features and one continuous target. The experiment compares Linear Regression, Ridge Regression and Lasso Regression using the supplied 80:20 train-test setup.

Property	Value
Samples	442
Features	10
Models	Linear, Ridge, Lasso
Best Ridge alpha	1.0
Best Lasso alpha	0.5

4. Theory

Overfitting occurs when a model learns the training data too closely and fails to generalize well to unseen observations. A model with excessive complexity may obtain a high training score but a noticeably lower test score.

Regularization controls model complexity by adding a penalty related to the magnitude of model coefficients. This discourages unnecessarily large coefficients and can improve generalization.

Ridge Regression uses L2 regularization. Its objective adds a penalty proportional to the sum of squared coefficients. Ridge usually shrinks coefficients toward zero but generally does not force them exactly to zero.

Lasso Regression uses L1 regularization. Its penalty is proportional to the sum of absolute coefficient values. Lasso can shrink some coefficients exactly to zero, providing a form of feature selection.

The regularization strength is controlled by alpha. A very small alpha behaves more like ordinary Linear Regression, while a larger alpha imposes a stronger penalty and can cause excessive shrinkage.

Cross-validation is useful for selecting alpha because it estimates performance on held-out folds of the training data. This helps avoid choosing a regularization parameter directly from the final test set.

Comparing train and test R^2 can provide evidence about generalization. If the training score is substantially higher than the test score, the model may be fitting patterns that do not transfer well to unseen data.

Regularization is particularly useful when features are correlated or when the number of predictors is large relative to the number of observations. It can stabilize coefficient estimates and reduce variance.

The Ridge objective can be represented conceptually as minimizing the residual sum of squares plus α times the sum of squared coefficients. The Lasso objective uses the sum of absolute coefficients instead. These different penalty shapes produce different coefficient behavior.

Standardization is commonly performed before regularization so that the penalty treats coefficients on a comparable scale. Without scaling, variables measured on different scales could receive unequal effective penalties.

The supplied experiment uses five-fold cross-validation on the training data to compare candidate alpha values. The best Ridge alpha is 1.0 and the best Lasso alpha is 0.5.

An important advantage of regularization is that the final test set can remain untouched during hyperparameter selection. After alpha is selected, the model can be evaluated once on the held-out test data to estimate generalization performance.

5. Methodology

- Load the Diabetes dataset.
- Split the data into training and testing sets.
- Standardize the features.
- Fit ordinary Linear Regression as a baseline.
- Evaluate Ridge over candidate alpha values using cross-validation.
- Evaluate Lasso over candidate alpha values using cross-validation.
- Select the best alpha for each regularized model.
- Compare train/test R^2, MAE, MSE and RMSE.
- Inspect coefficient shrinkage and residual behavior.

6. Outputs

6.1 Model Comparison

Model	Train R2	Test R2	Test MAE	Test MSE	Test RMSE
Linear Regression	0.5279	0.4526	42.7941	2900.1936	53.8534
Ridge Regression	0.5276	0.4541	42.8120	2892.0146	53.7775
Lasso Regression	0.5245	0.4607	42.8395	2857.4990	53.4556

6.2 Coefficient Comparison

Feature	Linear Regression	Ridge Regression	Lasso Regression
age	1.7538	1.8073	1.3423
sex	-11.5118	-11.4482	-10.3907
bmi	25.6071	25.7327	26.2262
bp	16.8289	16.7343	16.1689
s1	-44.4489	-34.6720	-11.6644
s2	24.6410	17.0531	0.0000
s3	7.6770	3.3699	-6.4309
s4	13.1388	11.7643	7.4560
s5	35.1612	31.3784	23.0849
s6	2.3514	2.4581	2.3296

6.3 Train vs Test R^2

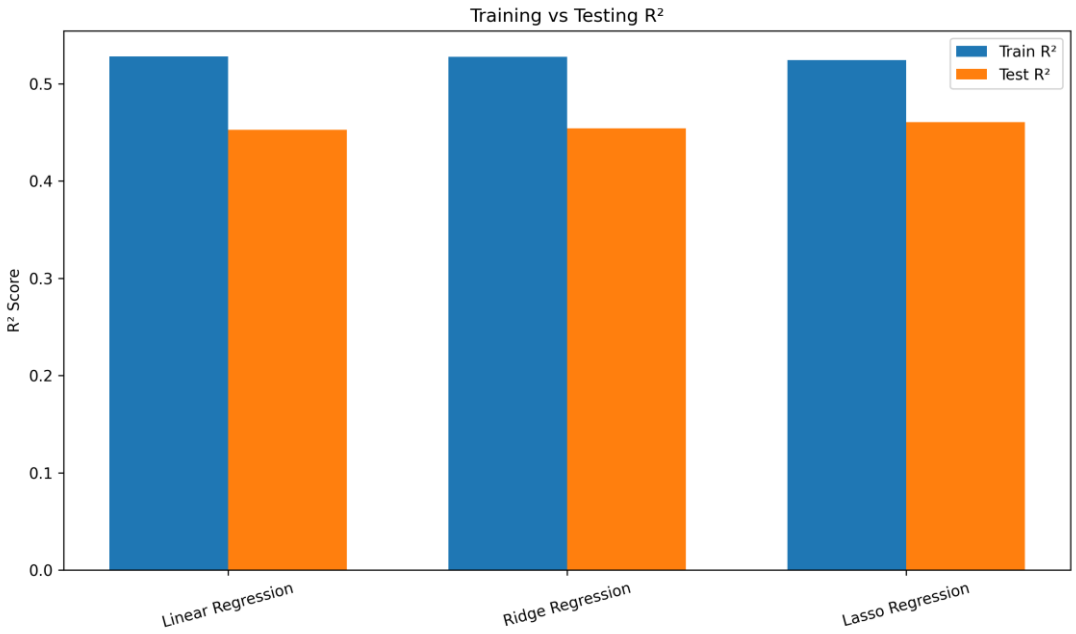


Figure 1: Training and testing R^2 comparison

6.4 Coefficient Comparison

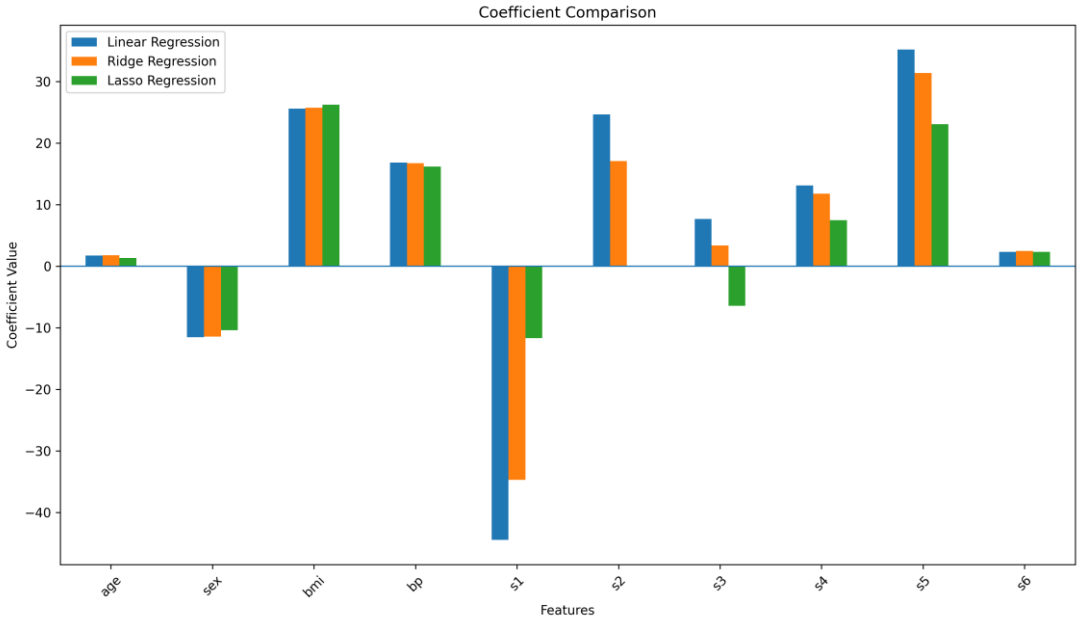


Figure 2: Coefficient comparison

6.5 Ridge Alpha vs CV MSE

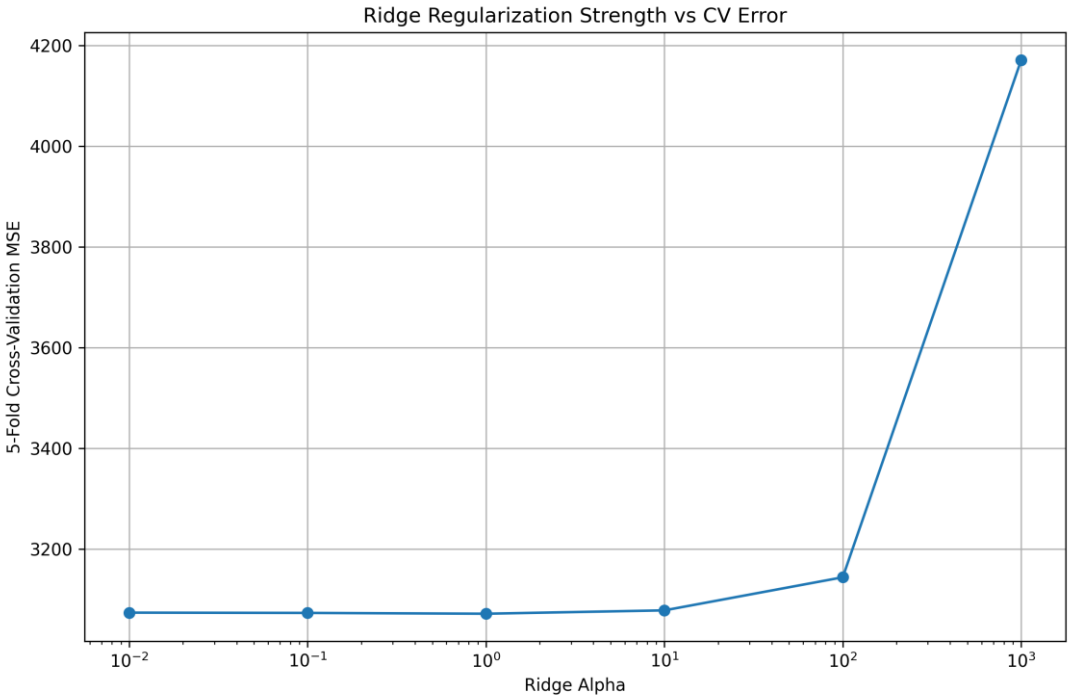


Figure 3: Ridge alpha versus cross-validation MSE

6.6 Lasso Alpha vs CV MSE

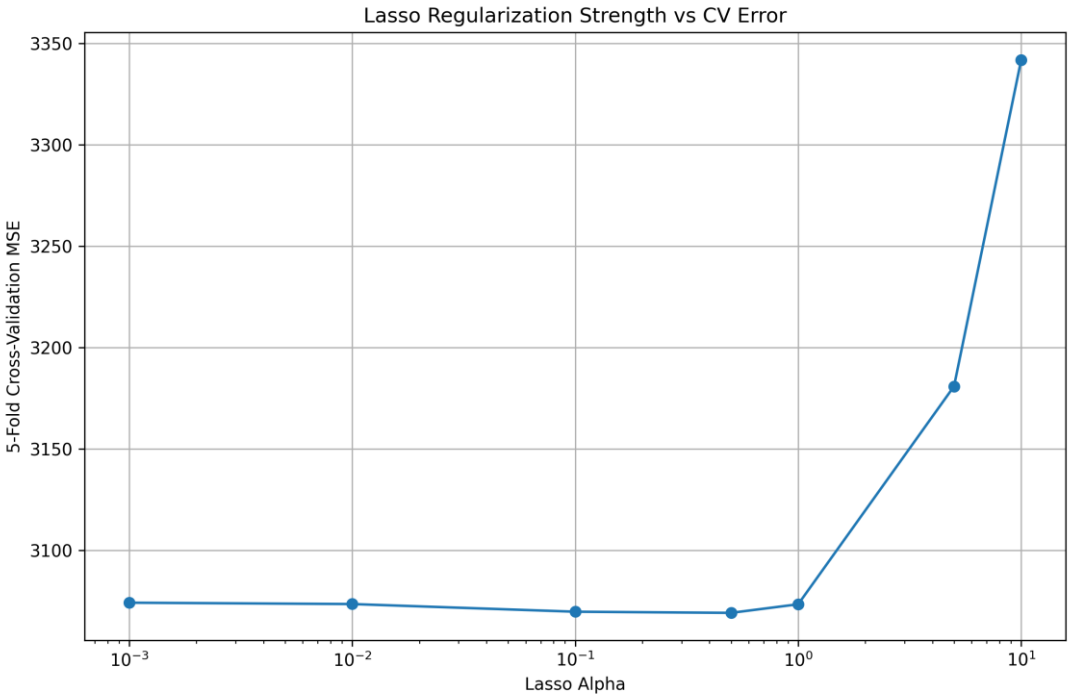


Figure 4: Lasso alpha versus cross-validation MSE

6.7 Actual vs Predicted and Residual Plot

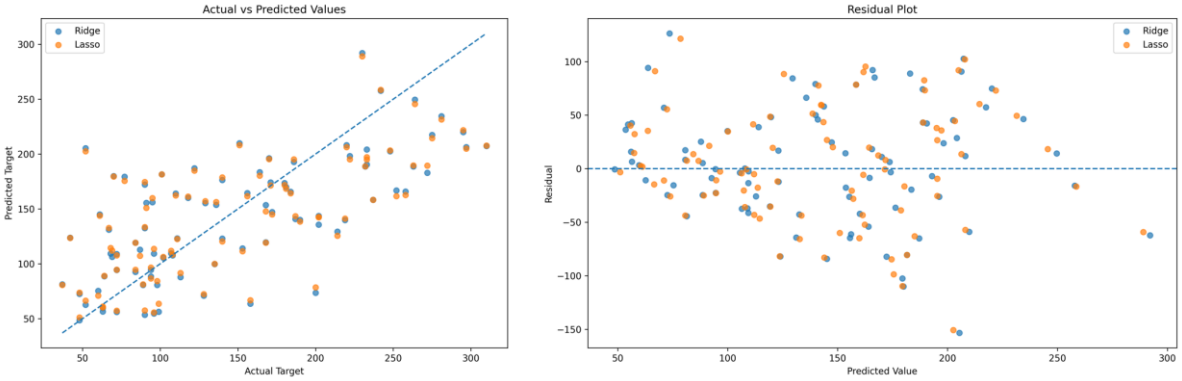


Figure 5: Actual versus predicted values (left) and residual plot (right)

7. Result

The supplied results show Linear Regression test $R^2 = 0.4526$, Ridge test $R^2 = 0.4541$ and Lasso test $R^2 = 0.4607$. Lasso produced the lowest test MSE (2857.4990) and RMSE (53.4556) among the three models. The selected alphas were 1.0 for Ridge and 0.5 for Lasso.

8. Observations

- Regularization slightly improved test performance relative to ordinary Linear Regression.
- Lasso reduced one coefficient to exactly zero in the supplied result.
- Ridge shrank coefficient magnitudes without setting coefficients exactly to zero.
- Cross-validation was used to select alpha on the training data.
- Lasso produced the best test R^2 among the three supplied models.

9. Conclusion

The experiment demonstrates how regularization can control model complexity and improve generalization. Ridge and Lasso provide different forms of coefficient shrinkage, and cross-validation provides a systematic method for selecting regularization strength.